Appendix C

Table C1: Records included in this scoping review that identified, measured, and valued environmental impacts in an economic context.

Author	Year	Title	Identified	Measured (Natural Units)	Valued
Adams et al.	2022	Environmental and financial impact of replacing a daily emergency anesthesia set-up with a mobile emergency case cart	Waste	kg	CO_2 using online calculators (Greenhouse Gas Reductions Calculator, Waste Reduction Model)
Adler et al.	2005	Comparison of economic and environmental impacts between disposable and reusable instruments used for laparoscopic cholecystectomy	Waste Water Energy Cleaning agents Steam Compressed air	kg m^3 kWh L kg m^3	-
Breth-Petersen et al.	2022	Health, financial and environmental impacts of unnecessary vitamin D testing: a triple bottom line assessment adapted for healthcare	Greenhouse gas emissions	No measurement in natural units, CO2-e obtained from literature.	CO_2-e
Büttner et al.	2020	Switching off for future—Cost estimate and a simple approach to improving the ecological footprint of radiological departments	Energy	kWh	CO_2-e
Carlassara et al.	2022	[Home Hemodialysis: new organizational models to favor the economic and environmental	Greenhouse gas emissions from	km	CO_2-e using calculator by MyClimate.

		sustainability in the province of Belluno]	patient and staff transport Waste		
Chu et al.	2021	Thinking green: modelling respirator reuse strategies to reduce cost and waste	Waste	g, kg	-
Connor et al.	2019	Clinical, fiscal and environmental benefits of a specialist-led virtual ureteric colic clinic: a prospective study	Greenhouse gas emissions from patient travel	km	CO_2-e using calculator by Carbon footprint
Croghan et al.	2021	Time, cost and carbon-efficiency: a silver lining of COVID era virtual urology clinics?	Greenhouse gas emissions from patient travel	miles	CO_2-e using calculator by Carbon footprint
Davie et al.	2010	Peripatetic intravenous service for metabolic bone disease: case study in patient centred-care for new NHS	Greenhouse gas emissions from patient travel	miles	CO_2 (assuming petrol fuel efficiency of 1 L per 9 miles and emissions of 2.3 kg per L petrol)
de Preux and Rizmie	2018	Beyond financial efficiency to support environmental sustainability in economic evaluations	Greenhouse gas emissions	No measurement in natural units, CO2-e obtained from literature.	CO2-e converted to monetary units (using UK non-traded price of carbon 2010; 52 GBP per ton CO_2-e)
Debaveye et al.	2018	The public health benefit and burden of mass drug administration programs in Vietnamese schoolchildren: Impact of mebendazole	Pharmaceutical supply chain	Range of natural units from Life cycle analysis	DALYs calculated using life cycle impact analysis.

Debaveye et al.	2019	Human health benefit and burden of the schizophrenia health care pathway in Belgium: paliperidone palmitate long-acting injections	Pharmaceutical supply chain	Range of natural units from Life cycle analysis	DALYs calculated using life cycle impact analysis.
Dullet et al.	2017	Impact of a University-Based Outpatient Telemedicine Program on Time Savings, Travel Costs, and Environmental Pollutants	Greenhouse gas emissions from patient travel	CO_2, VOC, CO	-
Eustachio Colombo et al.	2021	Pathways to "5-a-day": modeling the health impacts and environmental footprints of meeting the target for fruit and vegetable intake in the United Kingdom	Greenhouse gas emissions Water	No measurement in natural units, CO2-e obtained from literature. L (blue water footprint)	CO_2-e
Gatenby	2011	Modelling the carbon footprint of reflux control	Greenhouse gas emissions	No measurement in natural units, CO2-e obtained from literature.	CO_2-e
Goel et al.	2021	Improving productivity, costs and environmental impact in International Eye Health Services: Using the 'Eyefficiency' cataract surgical services auditing tool to assess the value of cataract surgical services	Waste Greenhouse gas emissions	Measured using "Eyefficiency app" and life cycle analysis	CO_2-e
Hensher	2020	Incorporating environmental impacts into the economic evaluation of health care systems:	Greenhouse gas emissions Particulate matter	-	Monetary units (converted from CO2-e using the social cost

		Perspectives from ecological economics	Waste Pharmaceutical contamination		of carbon. Examples provided from the UK, Canada, World Bank, IMF, and USA)
Kindred et al.	2024	Exploratory Approach to Incorporating Carbon Footprint in Health Technology Assessment (HTA) Modelling: Cost-Effectiveness Analysis of Health Interventions in the United Kingdom	Greenhouse gas emissions	-	CO_2-e, using NHS carbon intensity (0.156 kg CO_2-e / £)
Kponee-Shovein et al.	2022	Carbon footprint and associated costs of asthma exacerbation care among UK adults	Greenhouse gas emissions	No measurement in natural units, CO2-e obtained from literature.	CO2-e converted to monetary units (using UK non-traded price of carbon 2020; 71 GBP per ton CO_2-e)
Kponee-Shovein et al.	2022	Impact of choice of inhalers for asthma care on global carbon footprint and societal costs: a long-term economic evaluation	Greenhouse gas emissions	No measurement in natural units, CO2-e obtained from literature.	CO2-e converted to monetary units (using social cost of carbon from US Government Interagency Working Group on Social of Greenhouse Gases) CO2-e converted to DALYs (using conversions estimated by Tang et al.)

Lewis et al.	2009	Use of videoconferencing in Wales to reduce carbon dioxide emissions, travel costs and time	Greenhouse gas emissions from staff travel	km	CO_2 (using calculator published by the UK Government)
Marsh et al.	2016	Incorporating environmental outcomes into a health economic model	Greenhouse gas emissions		CO_2-e (estimated assuming average annual CO_2 intensity of pharmaceutical products purchased by the NHS)
Maughan et al.	2016	Cost and carbon burden of long-acting injections: a sustainable evaluation	Greenhouse gas emissions	-	CO_2-e
Maughan et al.	2016	Primary-care-based social prescribing for mental health: an analysis of financial and environmental sustainability	Greenhouse gas emissions	-	CO_2-e
Maughan et al.	2016	Evaluating sustainability: a retrospective cohort analysis of the Oxfordshire therapeutic community	Greenhouse gas emissions	-	CO_2-e
McGain et al.	2017	Financial and environmental costs of reusable and single-use anaesthetic equipment	Greenhouse gas emissions Water use Eutrophication Human, terrestrial, freshwater, marine ecotoxicity	Range of natural units from Life cycle analysis L Phosphorous deposition kg 1,4-DCB	CO_2-e

McGain et al.	2010	The Financial and Environmental Costs of Reusable and Single-Use Plastic Anaesthetic Drug Trays	Greenhouse gas emissions	Range of natural units from Life cycle analysis	CO_2-e
Miah et al.	2019	A prospective clinical, cost and environmental analysis of a clinician-led virtual urology clinic	Greenhouse gas emissions from patient travel	miles	CO_2-e using calculator by Carbon footprint
Michard et al.	2023	Pulse contour techniques for perioperative hemodynamic monitoring: A nationwide carbon footprint and cost estimation	Greenhouse gas emissions	No measurement in natural units, CO2-e obtained from literature.	CO_2-e
Minotti et al.	2022	True Cost Accounting of a healthy and sustainable diet in Italy	Greenhouse gas emissions	No measurement in natural units, CO2-e obtained from literature.	CO2-e converted to monetary units (converted from CO_2-e into Euros using the European ETS scheme carbon price, from Carbon Price Viewer)
Nardelli et al.	2023	Incremental peritoneal dialysis allows to reduce the time spent for dialysis, glucose exposure, economic cost, plastic waste and water consumption	Waste Water	kg L	-
Olmos et al.	2023	Evaluating the potential impact of spinal anesthesia use in lumbar surgery on global healthcare cost and climate change	Greenhouse gas emissions	No measurement in natural units, CO2-e obtained from literature.	CO_2-e

Ortsäter et al.	2020	Incorporating the Environmental Impact into a Budget Impact Analysis: The Example of Adopting RESPIMAT® Re-usable Inhaler	Greenhouse gas emissions	Range of natural units from Life cycle analysis	CO2-e converted to monetary units (converted from CO_2-e to dollars using a societal cost of carbon of 40 EUR per ton)
Ortsäter et al.	2019	A Budget Impact Model to Estimate the Environmental Impact of Adopting RESPIMAT® Re-usable in the Nordics and Benelux	Greenhouse gas emissions	Range of natural units from Life cycle analysis	CO2-e converted to monetary units (converted from CO_2-e to dollars using a societal cost of carbon of $50 USD per ton)
Papavero et al.	2023	Telemedicine Has a Social Impact: An Italian National Study for the Evaluation of the Cost-Opportunity for Patients and Caregivers and the Measurement of Carbon Emission Savings	Greenhouse gas emissions from patient travel	Km	CO_2-e using conversion from EU regulation
Paquette and Lin	2019	Outpatient Telemedicine Program in Vascular Surgery Reduces Patient Travel Time, Cost, and Environmental Pollutant Emissions	Greenhouse gas emissions from patient travel	CO_2, NOX, CO, VOC	-
Rey Aldana et al.	2021	Cost and Potential Savings of Electronic Consultation and Its Relationship with Reduction in Atmospheric Pollution	Greenhouse gas emissions from patient travel	Miles	CO2-e converted to monetary units (converted first into CO_2 and then into dollars using a

					conversion of EUR 0.0770235 per gram, previously estimated by Litman)
Rizan and Bhutta	2022	Environmental impact and life cycle financial cost of hybrid (reusable/single-use) instruments versus single-use equivalents in laparoscopic cholecystectomy	Greenhouse gas emissions Ozone depletion Ionising radiation Ozone formation Particulate matter Acidification Eutrophication Ecotoxicity Human toxicity Land use Mineral resources Fossil resource scarcity Water consumption	Range of natural units from Life cycle analysis	CO_2-e
Rouvière et al.	2022	Ecoresponsible actions in operating rooms: A health ecological and economic evaluation	Greenhouse gas emissions Occupied land Human toxicity Environmental toxicity Mineral resources Fossil resources Water consumption	Range of natural units from Life cycle analysis m^2 year crop eq kg 1,4-DCB kg 1,4-DCB kg Cu kg oil m^3 water	CO_2-e
Sanchez et al.	2020	Environmental and economic comparison of reusable and disposable blood pressure cuffs in multiple clinical settings	Greenhouse gas emissions	Range of natural units from Life cycle analysis	CO_2-e

Seconda et al.	2021	Conservative to disruptive diets for optimizing nutrition, environmental impacts and cost in French adults from the NutriNet-Santé cohort	Greenhouse gas emissions Energy Land occupation	Range of natural units from Life cycle analysis	The pReCiPe synthetic score (summarising environmental impact)
Spoyalo et al.	2023	Patient, hospital and environmental costs of unnecessary bloodwork: capturing the triple bottom line of inappropriate care in general surgery patients	Greenhouse gas emissions	Range of natural units from Life cycle analysis	CO_2-e
ten Have et al.	2022	Turning green: The impact of changing to more eco-friendly respiratory healthcare-a carbon and cost analysis of Dutch prescription data	Greenhouse gas emissions	No measurement in natural units, CO2-e obtained from literature.	CO_2-e
Tselapedi-Sekeitto et al.	2023	Telemedicine as an environmental ally - The social, financial, and environmental impact of virtual care in the otolaryngology clinic	Greenhouse gas emissions	km	CO_2-e
Unger and Landis	2016	Assessing the environmental, human health, and economic impacts of reprocessed medical devices in a Phoenix hospital's supply chain	Greenhouse gas emissions Human toxicity Particulate matter	Range of natural units from Life cycle analysis CTUh PM10e	CO_2-e

Vergunst et al.	2020	Applying the triple bottom line of sustainability to healthcare research—a feasibility study	Greenhouse gas emissions	No measurement in natural units, CO2-e obtained from literature.	CO_2-e
Vidal-Alaball et al.	2019	Impact of a Telemedicine Program on the Reduction in the Emission of Atmospheric Pollutants and Journeys by Road	Greenhouse gas emissions from patient travel	CO_2, NOX, CO, SO_2	-
Wilkinson et al.	2019	Costs of switching to low global warming potential inhalers. An economic and carbon footprint analysis of NHS prescription data in England	Greenhouse gas emissions	No measurement in natural units, CO2-e obtained from literature.	CO_2-e
Williams et al.	2023	Exploring the Integration of Environmental Impacts in the Cost Analysis of the Pilot MEL-SELF Trial of Patient-Led Melanoma Surveillance	Greenhouse gas emissions from patient travel	km	CO_2-e (using methods published by National Greenhouse Accounts Factors)

Table C2: Records included in this scoping review that discussed evaluating environmental impacts in an economic evaluation or HTA

Author	Year	Title	Method described	Method applied
de Preux and Rizmie	2018	Beyond financial efficiency to support environmental sustainability in economic evaluations	Cost utility analysis with environmental impacts converted from CO_2-e to dollars and included with other costs.	Yes, home haemodialysis vs in-centre haemodialysis
Debaveye et al.	2018	The public health benefit and burden of mass drug administration programs in Vietnamese schoolchildren: Impact of mebendazole	Markov model estimated the DALYs associated with the treatment effect and pharmaceutical supply chain was measured.	Yes
Debaveye et al.	2019	Human health benefit and burden of the schizophrenia health care pathway in Belgium: paliperidone palmitate long-acting injections	Markov model estimated the DALYs associated with the treatment effect and pharmaceutical supply chain was measured.	Yes
Greenwood Dufour et al.	2022	How We Might Further Integrate Considerations of Environmental Impact When Assessing the Value of Health Technologies	A discussion of ways that environmental impacts can be included in the deliberative HTA process, primarily via a parallel evaluation	No
Hensher	2020	Incorporating environmental impacts into the economic evaluation of health care	A discussion of ways that environmental impacts can be included in economic evaluations, primarily including the impacts as a cost in CEA and CUA	Yes, using the previously published example by Marsh et al. of adding insulin to an oral antidiabetic medication treatment regime

		systems: Perspectives from ecological economics		
Kindred et al.	2024	Exploratory Approach to Incorporating Carbon Footprint in Health Technology Assessment (HTA) Modelling: Cost-Effectiveness Analysis of Health Interventions in the United Kingdom	Proposes the incremental carbon footprint effectiveness ratio (ICFER) and the incremental carbon footprint cost ratio (ICFCR) to report alongside the ICER in cost effectiveness studies.	Yes, using the example of semaglutide and bariatric surgery for the treatment of obesity
Marsh et al.	2016	Expanding Health Technology Assessments to Include Effects on the Environment	A discussion about the approaches and challenges of various methods for including environemtnal impacts in HTA, specifically: Converting impacts to health units and including them on the benefits side of a CUA or CEA – "Enriched CUA" Adjust WTP thresholds of CEA/CUA to reflect environmental impacts Environmental impacts can be converted into dollars and included alongside other benefits in a CBA Environmental impacts can be included as a criteria of a MCDA	No
Marsh et al.	2016	Incorporating environmental outcomes into a health economic model	Environmental impacts included in an cost utility model (IMS CORE diabetes model). Model estimated impacts but did not include them in an ICER or further evaluation.	Yes, adding insulin to an oral antidiabetic medication treatment regime

McAlister et al.	2022	Incorporating carbon into health care: adding carbon emissions to health technology assessments	Environmental impacts can be used during HTA deliberations as a simple modifier when health outcomes and costs are equal Environmental impacts can be included as a criteria of a MCDA Environmental impacts can be converted into dollars and included alongside other benefits in a CBA Included as a cost in a CEA or CUA	No
Toolan et al.	2023	Environmental impact assessment in health technology assessment: Principles, approaches, and challenges	Proposes four approaches for the inclusion of environmental impacts in HTA: Information conduit Integrated evaluation Parallel evaluation Environment-focused evaluation	No
Walpole et al.	2023	How can environmental impacts be incorporated in health technology assessment, and how impactful would this be?	A discussion about the incorporation of environmental impacts in HTA, with methods discussed including: Environmental impacts could be integrated into the economic cost-effectiveness analysis Environmental impacts could be included in HTA deliberations	No